PROJECT DURATION
2 Months
TEAM MEMBERS – 4
Meghana, vineetha, praveen, charan
Role: Team Leader – Technical Research & Electrical System Integration

SCARA Robot with an Antistatic Coating Check Station for Quality Control

品質管理用帯電防止コーティング検査ステーション付きSCARAロボット



BUILT PROTOTYPE

SCARA arm, Z-column, stepper drives

Problem Statement / 問題提起

Slow speed → **Parts jamming** → **Large footprint** → **Manual coating checks: slow, operator-dependent**

Objectives / 目的

High Speed
Short, direct moves with two rotary joints

Flexible XY Axis
Compliant motion tolerates misalignment

Small Footprint
Compact base with vertical Z-column

PROPOSED FEATURE

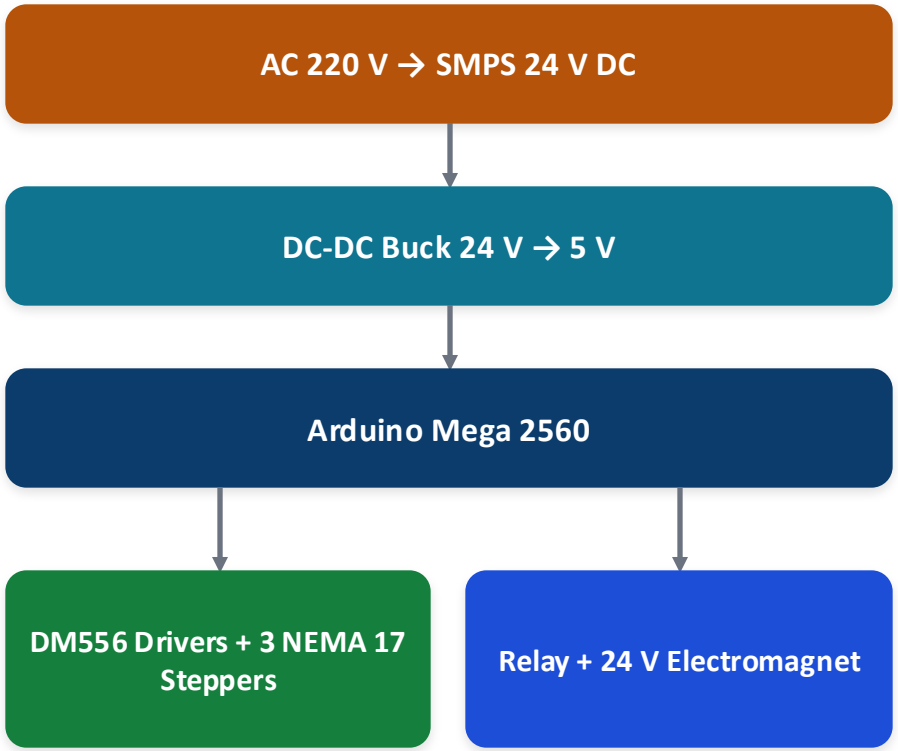
Inspection Task
Antistatic coating check station

SYSTEM ARCHITECTURE & METHODOLOGY

システム構成と手法

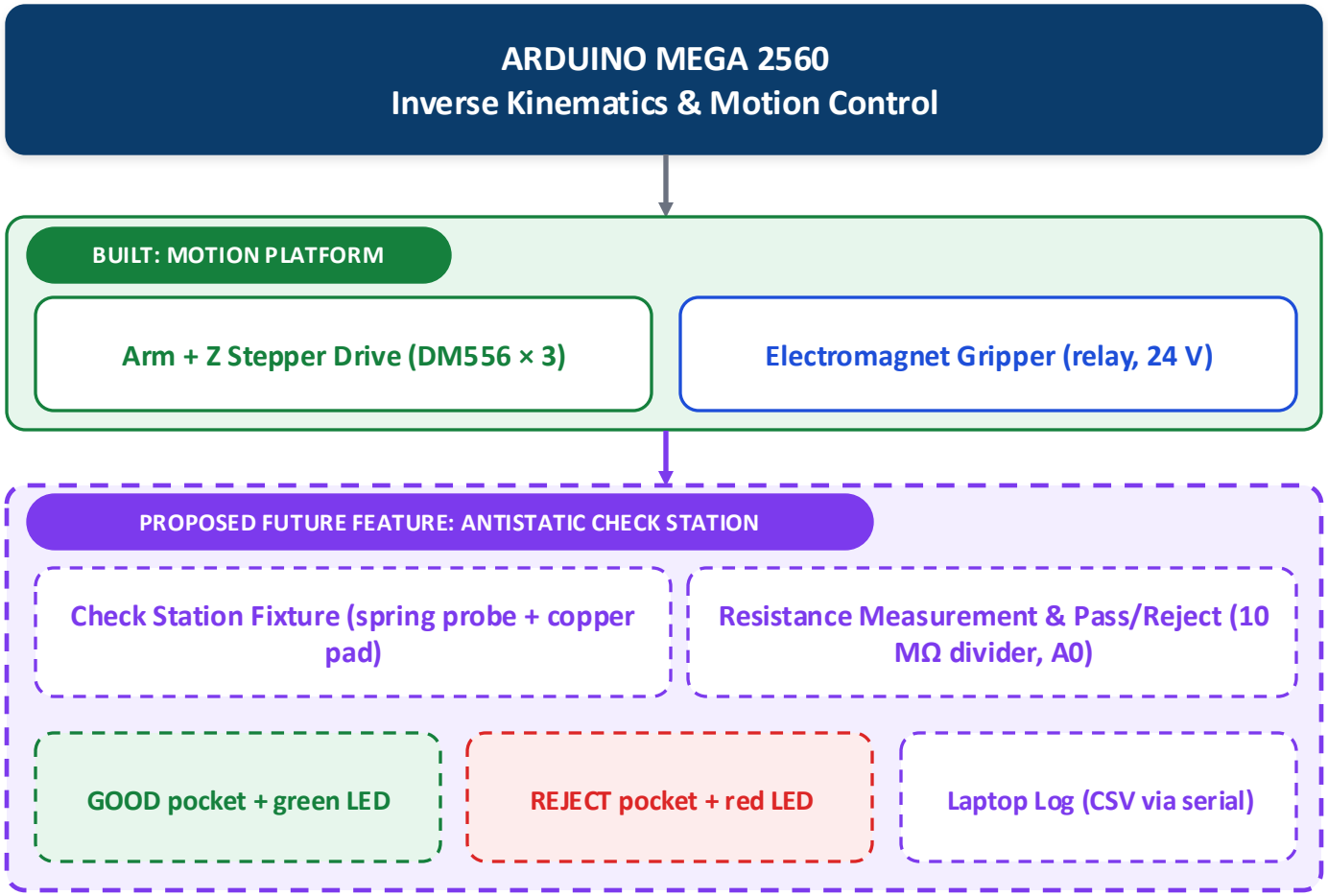
Hardware Architecture

BUILT



Arduino Mega computes inverse kinematics and sends step/direction pulses to each driver.

System Architecture & Flow / 動作原理



QUALITY CONTROL

品質管理

Check-station pass / reject window (design limits: R_LOW = 100 kΩ, R_HIGH = 1 GΩ)

PROPOSED

REJECT
R < R_LOW: bare metal or breached coating

GOOD
R_LOW ≤ R ≤ R_HIGH: antistatic coating in specification

REJECT
R > R_HIGH: insulating coating or no contact

Inspection & Test Plan

Item	Check / tool	Acceptance
Printed PETG parts	Dimensions and fit; calliper	Matches drawing SRP-001
Wiring	Continuity, polarity; multimeter	Traced and labelled
Motion	Homing repeatability, drift	≤ ±0.5 mm; ≤ 1 mm drift
Electromagnet	Pick and release; test coupons	Lifts and releases all
Check station (proposed)	Reference coupons, 3 classes	Clear guard band

Process Control

Start-of-session checks
Home, empty cycle, verify Z height, test 3 reference coupons.

Error-proofing
Probe arm lifted by default; red and green LEDs never on together.

Corrective action
5-Why and PDCA; one change at a time, recorded.

DATA ANALYSIS

データ解析

Data Logged per Cycle

- Cycle number and coupon class
- ADC reading and resistance R_x
- Decision: GOOD or REJECT
- Pick confirmed: yes / no
- Cycle time
- Saved as CSV from the serial port

PLANNED WITH CHECK STATION

Analyses

Analysis	Used for
Mean ± 3σ	Repeatability of position and readings
Limit setting	R_LOW, R_HIGH midway (log scale) between classes
Confusion matrix	Accuracy and false-accept rate
Control chart	Drift of the measurement system
Cycle-time breakdown	Find the bottleneck (Z, XY, dwell)
Pareto of faults	Fix the biggest cause first
Cpk	Is accuracy good enough for the task

Targets

≥ 99%
Pick success

≥ 99%
Classification accuracy

0
False accepts

±0.5 mm
XY repeatability

≤ 8 s
Cycle per coupon

≤ 1 mm
Home drift

Targets are acceptance criteria, to be confirmed by the 100-coupon validation run. No measured station results are claimed yet.

PROJECT OUTCOMES & RESULTS

結果と今後の展望

Outcomes

BUILT



Working SCARA robot: 3-axis motion with electromagnet gripper



Tested speeds: 150 mm/s (X-Y) and 60 mm/s (Z)

PROPOSED

Antistatic check station is the next stage: fixture, measurement and logging not built yet

Future Scope

PROPOSED

- **Servo-lift probe arm and resistance circuit**
- **100-coupon validation run (target 99%)**
- **Control chart and Cpk on real data**
- **Joint encoders and vision integration**

Performance Calculations

Metric	Basis	Result
Horizontal (X-Y) speed	Tested	**150 mm/s**
Vertical (Z) speed	Tested	**60 mm/s**
Reach	L1 + L2 = 160 + 120 mm	280 mm
Tip resolution at full reach	Calculated, worst case	≈ 0.8 mm
Inspection cycle per coupon (proposed station)	Calculated from tested speeds (25 / 10 mm Z stroke)	9.3 s / 7.3 s
Handling + classification efficiency (proposed)	100-coupon validation run	Target 99%

Thank You

ありがとうございます

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Team of 4 • Team Leader

